

Team Worksheet — Lab 17

Hypothesis Testing — In Depth, With Test Selection

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-17-hypothesis-testing.xlsx`.

3. What we produced

Deliverable: Completed hypothesis tests on your capstone's candidate causes with documented conclusions.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap the logic of hypothesis testing in full: assume H_0 (no difference), then ask how likely the observed data would be if H_0 were true.

Step 2. Recap H0 and Ha formulation. H0 always contains the equality; Ha carries the claim you are trying to establish.

Step 3. Recap alpha, beta and power. Alpha is producer's risk (Type I, false alarm); beta is consumer's risk (Type II, missed signal); power = 1 - beta.

Step 4. Understand the practical consequence: an underpowered test that fails to reject H0 has proven nothing. Check power before concluding 'no difference'.

Step 5. Learn the test selection logic in full. Continuous Y: 1 sample vs target = 1-sample t; 2 independent groups = 2-sample t; paired = paired t; 3+ groups = ANOVA; variances = F-test or Levene.

Step 6. Learn the discrete-Y branch: 1 proportion = 1-proportion test; 2 proportions = 2-proportion test; categorical association = chi-square.

Step 7. Check the assumptions before every test: normality, independence and equal variance. A violated assumption invalidates the p-value entirely.

Step 8. Take your five candidate causes from Lab 16 and select the correct test for each. Write down the justification for each selection.

Step 9. State H0 and Ha for each test, set alpha (normally 0.05), and confirm you have adequate sample size and power.

Step 10. Run each test and record the p-value. Apply the decision rule: $p < \alpha$ rejects H0.

Step 11. Distinguish statistical from PRACTICAL significance. A p-value of 0.001 on a 0.2 percent improvement is real but worthless — always report effect size alongside p.

Step 12. Build your rejected-causes log so nobody re-litigates cleared causes in Improve, and record confirmed causes in your capstone pack.

7. Self-check

Every test has a documented selection justification and assumption check, results report effect size alongside the p-value, and you maintain both a confirmed-cause and a rejected-cause log.

Complete? ■ Yes ■ Not yet

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